

CHARLES (CHUCK) FORSYTH

Research Computing and AI Executive | HPC | Research Cloud | Navy Veteran

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SUMMARY

Research computing and AI executive with 30+ years from Navy radar to HPC to generative AI. Director of Research Computing at UC Riverside, leading a matrix team of about 20 that serves about 1,000 researchers in 22 colleges and schools. Led UCR's first-of-its-kind Google Cloud research partnership, Co-PI on a \$498K NSF storage award, past Chair of the UC Research IT Committee. Building with AI since GPT-2 (2020); I build the AI platforms myself, with a team of AI agents.

HIGHLIGHTS

- **Google Cloud partnership:** led Ursa Major, announced by Google in 2023 as a first-of-its-kind research subscription; 170+ research projects under one governed, budgeted cloud.
- **NSF award:** Co-PI on CC* award 2346636 (\$498,553), which delivered 2.09 PB of research storage on the 100 Gbps Science DMZ, 422 TB shared with the National Research Platform.
- **AI for research:** built UCR's governed AI stack (MCP servers, AI Gateway, elastic GPU cluster) and ship it with AI agents: 240+ releases across ten tools in five weeks.
- **Campus and systemwide leadership:** founded and lead UCR Research IT Leaders; Research Advisory Board and Academic Senate research committee; Chair of the UC Research IT Committee (2022-2024), uniting research IT leaders from every UC campus.

EXPERIENCE

Director of Research Computing

Mar 2023 - Present

University of California, Riverside

- Lead research computing for about 1,000 researchers through a matrix team of about 20 (direct reports, the ITS systems team and campus research IT staff on dotted lines).
- Founded and lead UCR Research IT Leaders; Research Advisory Board; Academic Senate research committee; official UCR PI status (2024).
- Built UCR's AI-for-research stack: MCP servers with sign-in, roles and audit; the UCR AI Gateway with per-lab budgets; an elastic GPU cluster on Google Cloud.
- Built Nexus, the research knowledge graph and MCP server Research Computing runs on (943 researchers, 176 cloud projects).
- Data security plans and secure enclaves for P3/P4 and CMMC research with the Information Security Office.

Associate Director of Research Computing

Aug 2019 - Mar 2023

University of California, Riverside

- Grant strategy and budgets for research technology; lead systems architect; proxy for the CIO and CTO with the academic senate.

Chair, UC Research IT Committee

Jul 2022 - Jan 2024

University of California (systemwide)

- Led research IT leaders from every UC campus; Vice Chair Jun 2021 - Jul 2022.

Lead Linux & HPC Systems Administrator / Consultant

Jun 2017 - Aug 2019

UC Riverside, High Performance Computing Center (HPCC)

- Rebuilt the HPCC cluster and its server room, power and cooling; Ansible and PXE provisioning; built the HPCC Cloud Service (on-demand AWS clusters); taught 19 HPCC workshops.

Lead Linux & HPC Systems Administrator / Research Consultant

Jan 2012 - Jun 2017

Corning Incorporated

- Lead HPC architect for R&D: thousands of cores, Lustre, InfiniBand, 60+ licensed science codes.
- **Earlier:** owner, CnL Networks (2003-2012); Hancock Telephone; IBM; AOL; U.S. Navy electronics technician, USS Lake Champlain (1994-1997).

EDUCATION AND SKILLS

- B.S., Information Technology, Purdue University Global (GPA 3.89). Certifications: PBS Professional, Mellanox InfiniBand, EMC Isilon, PMI Project Management Foundations.
- AI and MCP, HPC and Slurm, Google Cloud and AWS, Kubernetes, Ceph and Lustre, NIST 800-171 and CMMC, Python and Go.